

Skill 6

Kodam Sathwika – 2520090210

JWT

Installation:

```
(.venv) PS C:\FastAPI_K1> python -m pip install python-jose
Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading python_jose-3.5.0-py2.py3-none-any.whl (34 kB)
Downloading rsa-4.9.1-py3-none-any.whl (34 kB)
Downloading ecdsa-0.19.2-py2.py3-none-any.whl (150 kB)
Downloading pyasn1-0.6.4-py3-none-any.whl (84 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: six, pyasn1, rsa, ecdsa, python-jose
Successfully installed ecdsa-0.19.2 pyasn1-0.6.4 python-jose-3.5.0 rsa-4.9.1 six-1.17.0
(.venv) PS C:\FastAPI_K1> python -m pip install python-multipart
Collecting python-multipart
  Downloading python_multipart-0.0.32-py3-none-any.whl.metadata (2.1 kB)
  Downloading python_multipart-0.0.32-py3-none-any.whl (30 kB)
Installing collected packages: python-multipart
Successfully installed python-multipart-0.0.32
```

Code:

```
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4 app = FastAPI()
5 SECRET_KEY = "my secret key"
6 ALGORITHM = "HS256"
7 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="/login")
8 @app.post("/login")
9 def login(form_data: OAuth2PasswordRequestForm = Depends()):
10     if form_data.username == "101" and form_data.password == "abc123":
11         payload = {
12             "sub": form_data.username,
13             "role": "student"
14         }
15
16         token = jwt.encode(
17             payload,
18             SECRET_KEY,
19             algorithm=ALGORITHM
20         )
21
22         return {
23             "access_token": token,
24             "token_type": "bearer"
25         }
26
27     raise HTTPException(
28         status_code=401,
29         detail="Invalid username or password"
30     )
31
32 def get_current_user(token: str = Depends(oauth2_scheme)):
```

```
def get_current_user(token: str = Depends(oauth2_scheme)):
    try:
        payload = jwt.decode(
            token,
            SECRET_KEY,
            algorithms=[ALGORITHM]
        )
        return payload

    except Exception:
        raise HTTPException(
            status_code=401,
            detail="Invalid or expired token"
        )

@app.get("/profile")
def profile(user=Depends(get_current_user)):
    return {
        "message": "Welcome to your profile",
        "user_id": user["sub"],
        "role": user["role"]
    }
```

default

GET /students Get Students

Parameters

No parameters

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls Accept header. Example Value Schema	No links